Olivia Kaitlyn Roberts, MS – Curriculum Vitae

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EDUCATION

University of Michigan August 2024 – May 2026
Master of Science in Computational Epidemiology and Systems Modeling GPA: 3.9
Certificate in Complex Systems

University of South Dakota August 2021 – May 2024
Bachelor of Science in Mathematics, Summa Cum Laude GPA: 3.9
Minors in Music and Interdisciplinary Sciences

RESEARCH EXPERIENCE

University of Michigan School of Public Health, Center for the Assessment of Tobacco Regulations (CAsToR)

Epidemiology Research Associate October 2026 – Present

Epidemiology Research Assistant May 2025 – October 2026

- Conducts research under the mentorship of Dr. Andrew Brouwer at CAsToR.
- Applies multistate transition models and latent class analysis using complex survey weights in R to analyze longitudinal trends in tobacco product use among adults and youth in nationally representative studies.
- Manages multiple research projects, conducts statistical analyses, prepares scientific manuscripts for publication, and presents research findings at academic conferences.
- Progressed from a full-time summer internship (May – August 2025) to part-time research during the academic year, returned to full-time research in May 2026, and was promoted to Research Associate in October 2026.

UDiscover Summer Research Scholar, University of South Dakota May 2023 – August 2023

- Conducted pure mathematics research with Dr. Gabriel Picioroaga on generalizing music theory using algebra.

Behavioral Health Epidemiology Intern, South Dakota Department of Social Services May 2022 – August 2022

- Analyzed state-level data on substance use and mental health using R, Excel, and Power BI.
- Contributed to behavioral health data and outcomes reports for state government and public use.

PUBLICATIONS

- Brouwer AF, **Roberts OK**, Jeon J, Jimenez-Mendoza E, Land SR, Freedman ND, Torres-Alvarez R, Mistry R, Levy DT, Meza R. Patterns of cigarette and electronic nicotine delivery system transitions in U.S. adults, 2021–2024. *American Journal of Preventative Medicine*. 2026;108576. doi:10.1016/j.amepre.2026.108576.
- Picioroaga G, **Roberts O**. Musical systems with $\mathbb{Z}_n$ – Cayley graphs. *MusMat: Brazilian Journal of Music and Mathematics.* 2024;8(1). doi:10.46926/musmat.2024v8.1-23.

TECHNICAL SKILLS

- Programming and Statistical Software: R, Python, Matlab, Stata, SQL
- Statistical and Computational Modeling: Markov multistate transition models, network modeling, deterministic and stochastic infectious and chronic disease modeling, latent class analysis
- Tools: Microsoft Excel, Word, PowerPoint, Power BI, SEER Stat, LaTeX

MANUSCRIPTS IN PROGRESS

Under Review

- **Roberts OK**, Bover Manderski MT, Gonsalves NJ, Jeon J, Meza R, Delnevo CD, Brouwer AF. Trends in transitions in cigarette and electronic nicotine vapor product use in the Rutgers Omnibus Study, 2022–2026. *Manuscript submitted to Nicotine & Tobacco Research.*
- **Roberts OK**, Sanchez-Romero LM, Jimenez-Mendoza E, Jeon J, Travis N, Mistry R, Meza R, Levy DT, Brouwer AF. Rates of oral nicotine pouch experimentation, initiation, and cessation by other tobacco product use and sociodemographic characteristics among U.S. adults, 2022–2024. *Manuscript submitted to Addiction.*

In Preparation

- **Roberts OK**, Jeon J, Jimenez-Mendoza E, Land SR, Freedman ND, Torres-Alvarez R, Mistry R, Levy DT, Meza R, Brouwer AF. Patterns of cigarette and electronic nicotine delivery system transitions in US youth, 2015–2024.
- Van Wyk H, **Roberts OK**. Dengue virus transmission patterns at fine spatial scales: human movement patterns can describe dengue outbreak dynamics in northwestern Ecuador.

PRESENTATIONS

"Trends in transitions in cigarette and electronic nicotine vapor product use in the Rutgers Omnibus study, 2022–2025"

- Poster presentation, NIH Tobacco Regulatory Science Meeting, July 2026
- Poster presentation, University of Michigan Epidemiology Poster Session, April 2026

"Transitions in cigarette, ENDS, and dual use among youth in the PATH Study from 2015–2023: a multistate transition modeling analysis"

- Invited seminar talk (1 hour), University of South Dakota Department of Mathematics, March 2026
- Oral presentation, Society for Research on Nicotine and Tobacco (SRNT) Meeting, March 2026

"Musical Systems with $\mathbb{Z}_n$ – Cayley Graphs"

- Oral presentation, Joint Mathematics Meetings (JMM) Pi Mu Epsilon Session, January 2024
- Oral presentation, Nebraska Conference for Undergraduate Women in Mathematics (NCUWM), January 2024
- Oral presentation, National Conference on Undergraduate Research (NCUR), April 2023 and April 2024

AWARDS

Research Awards and Grants

- Pi Mu Epsilon Outstanding Student Speaker Award, Joint Mathematics Meetings, January 2024
- Undergraduate Research Excellence Award, University of South Dakota, September 2023
- UDiscover Summer Research Grant, University of South Dakota, May 2023
- John W. Carlson Research Grant, University of South Dakota, April 2023

Scholarships

- Epidemiology Student Scholarship, University of Michigan, February 2024 (75% of tuition)
- South Dakota NASA Space Grant Consortium Educational Stipend, October 2023
- Dr. Cleland Cook Mathematics Scholarship, University of South Dakota, April 2022 and April 2023

TEACHING EXPERIENCE

Yoga Teacher, University of Michigan Recreation — August 2024 – Present

- Instructs yoga classes and developed a 6-week Elevate Yoga workshop series to deepen students' yoga practice.

Instructor Aide, University of Michigan Department of Mathematics — July 2024 and July 2025

- Co-instructed three summer camp sessions for 15-20 high school students for the Michigan Math and Science Scholars program.
- Assisted with curriculum development and delivered lectures on group theory, set theory, and music theory.

Instructor Assistant, University of Michigan School of Nursing — January 2025 – May 2025

- Supported 100 students in NURS 218 (Applied Statistics).
- Held office hours, graded assignments, assisted students with Stata programming, and led review sessions.

Math Tutor, University of South Dakota — January 2022 – May 2024

- Tutored undergraduate students in algebra, statistics, and calculus.

Yoga Teacher, University of South Dakota Wellness Center — August 2021 – May 2024

- Instructed yoga classes and workshops and organized the first and second annual USD Yoga Festivals.

OTHER EXPERIENCE

Event Volunteer, VegMichigan — June 2024 – Present

CPR/AED for Professional Rescuers Certification — August 2026

U-M Summer Session, Simulation Modeling of Tobacco Use, Health Effects and Policy Impacts — July 2026

Admissions Ambassador Volunteer, University of Michigan School of Public Health — August 2024 – May 2026

200-Hour Yoga Teaching Certification, Tiny Buddha Yoga, Ann Arbor, MI — December 2024

200-Hour Yoga Teaching Certification, Better Together Yoga, Sioux Falls, SD — September 2021